

Prerequisites

You must have completed the Min Heap Home Assignment and bring your working MinHeap.h implementation.

Problem Statement

Add a new function *kLargest(int k, std::ofstream &outfile)* to your MinHeap class that returns the k largest elements from the heap following the below restrictions

- The function must return elements in sorted (descending) order
- The original heap must NOT be modified after the operation
- Time complexity should be $O((n+k)\log n)$

Command	Operation	Description
13 k	kLargest(k)	Prints the K largest elements in the heap

Sample I/O

Sample Input	Sample output
10 7 50 30 20 15 10 40 25 35 8 13 3 8 13 5	Heap built from array. Heap: 10 15 20 25 35 50 40 30 K Largest: 50 40 35 Heap: 10 15 20 25 35 50 40 30 K Largest: 50 40 35 30 25